

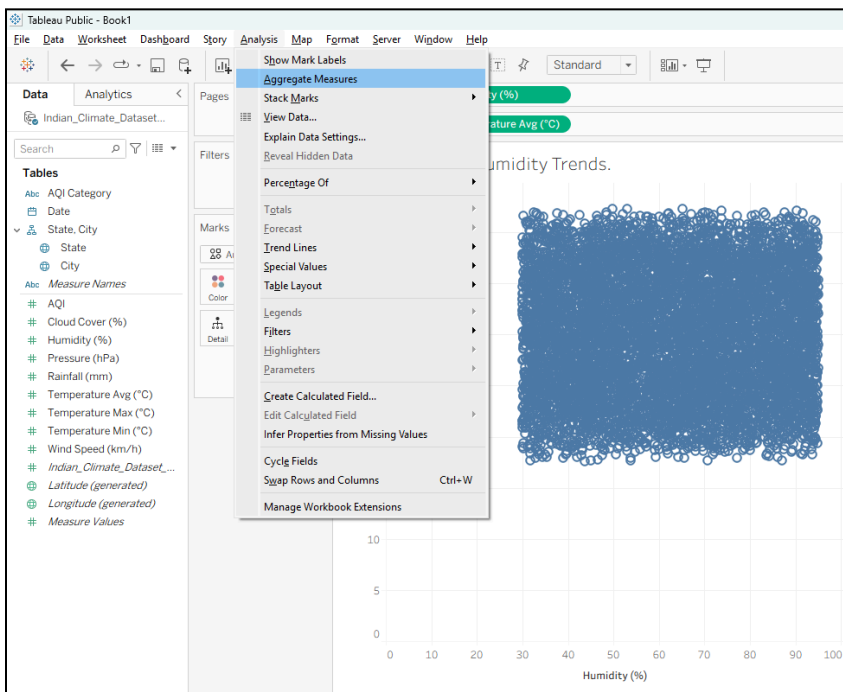
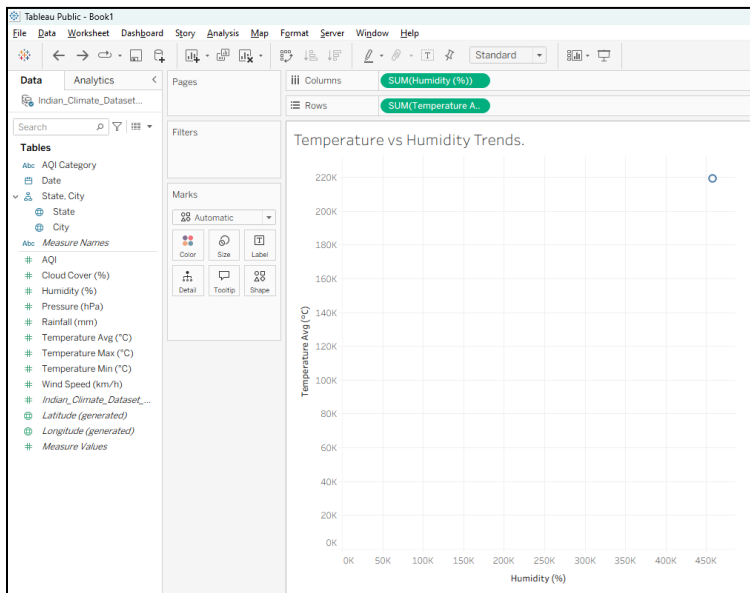
Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – VIII

Roll No.:T044	Name:SACIHN SINGH
Class:TY.IT	Batch:1
Date of Assignment:27/08/2026	Date/Time of Submission:27/08/2026

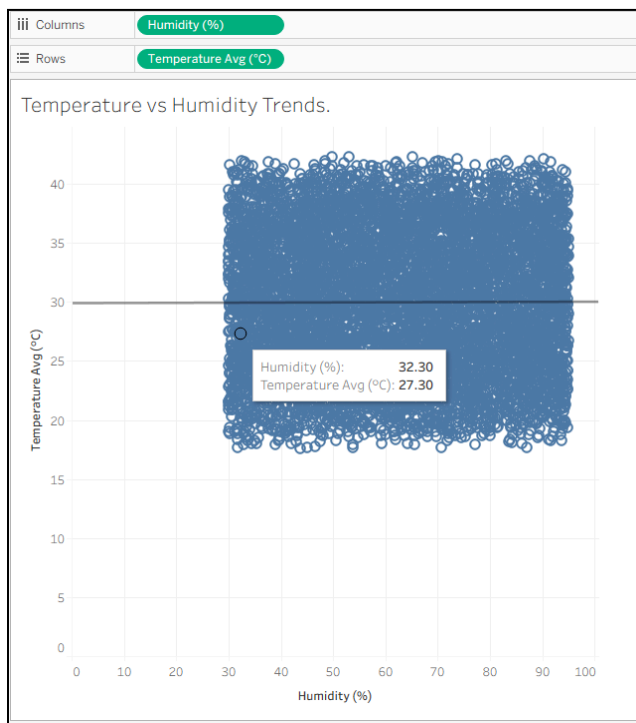
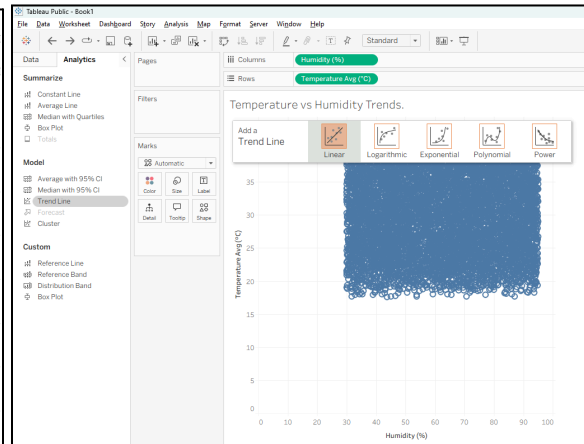
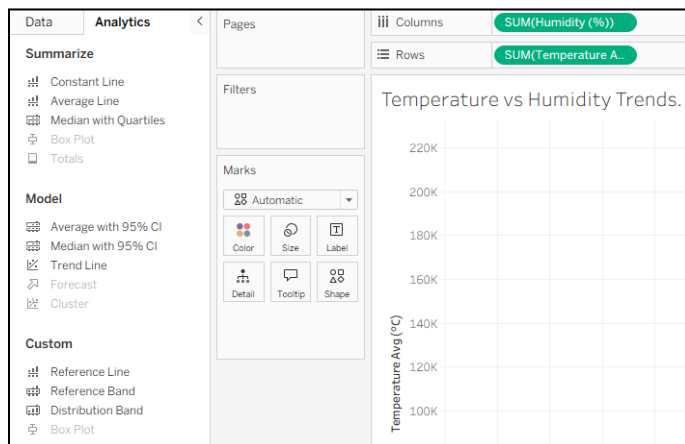
a) Add and interpret trend lines (linear, exponential)

1. Connect Dataset & Build a Scatter Plot / Line Chart

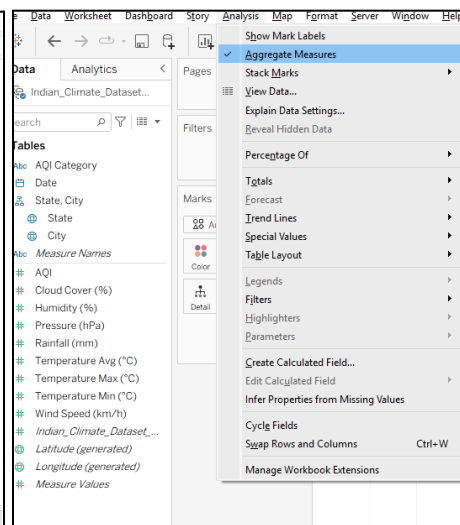
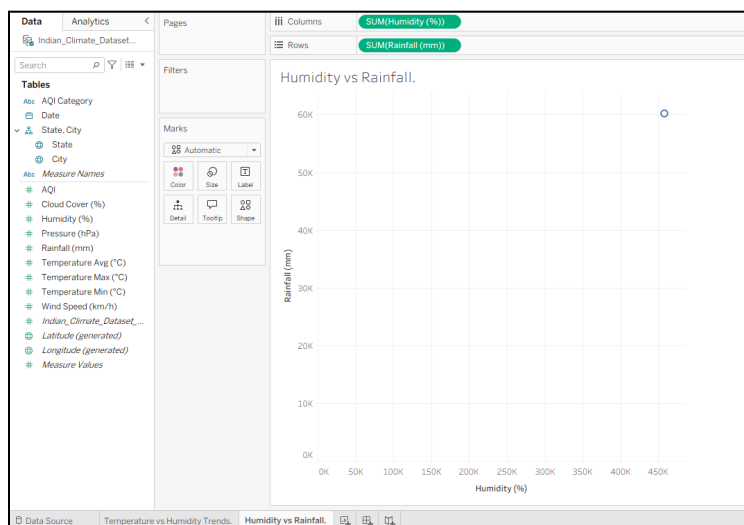


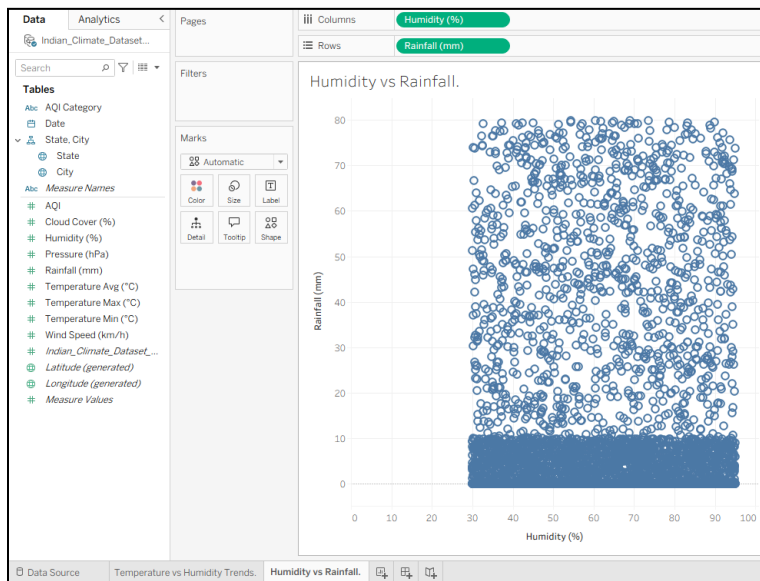
2. Add a Linear Trend Line

Switch from Data to Analytics



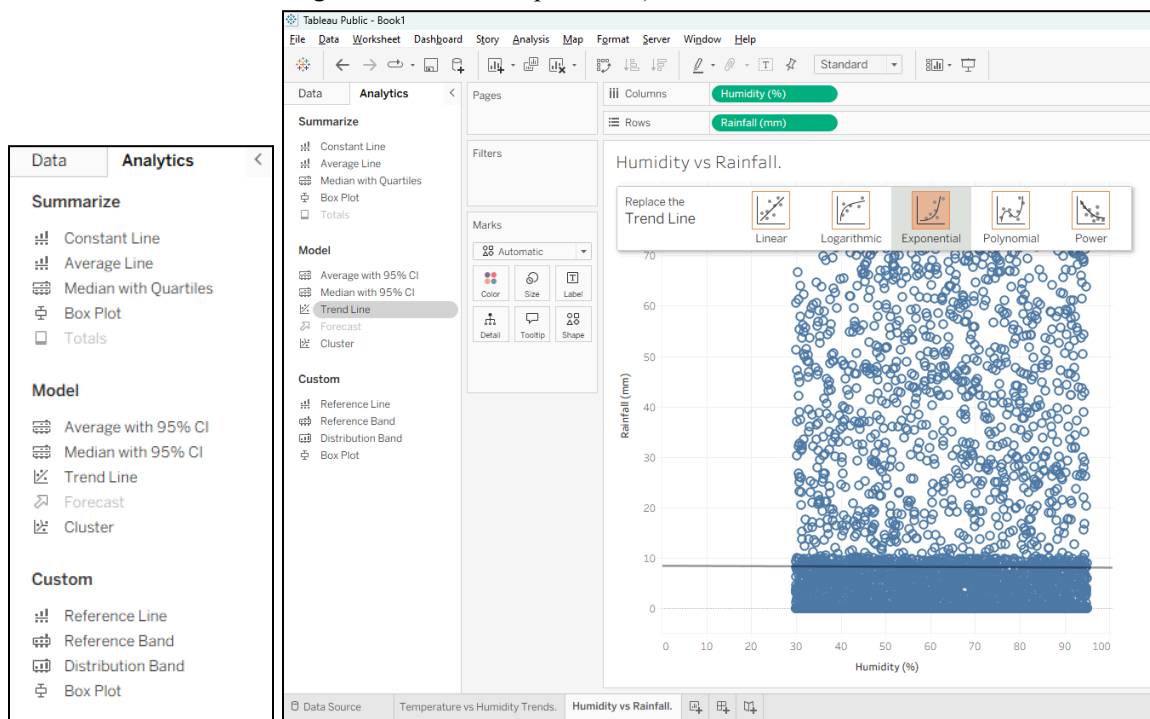
3. Add an Exponential Trend Line for Non-Linear Measures (Rainfall vs. Humidity)





Switch from **Data** to **Analytics**

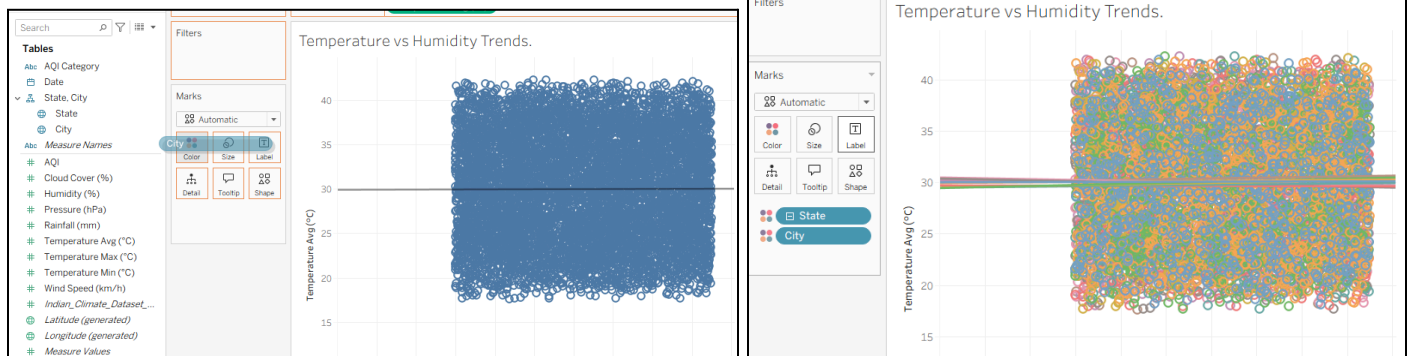
From the Analytics pane, drag Trend Line onto Exponential (or right-click the canvas Trend Lines Show Trend Lines Edit Trend Lines and change the model to Exponential).



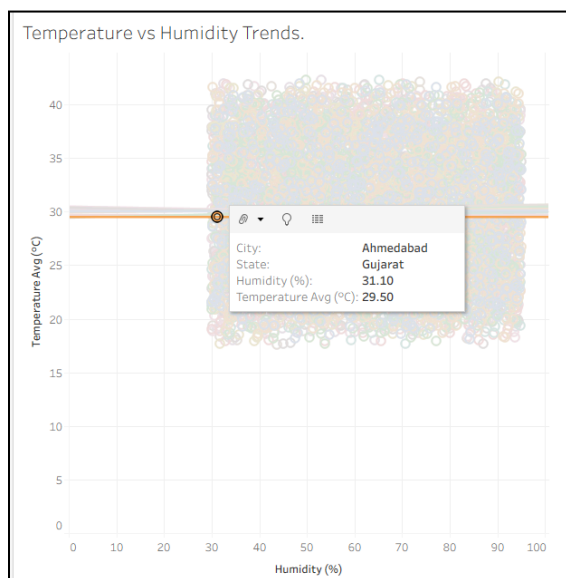
b) Customize and analyze trend models

1. Disaggregate Trend Lines by City

Drag City from the Data pane onto Color on the Marks card.

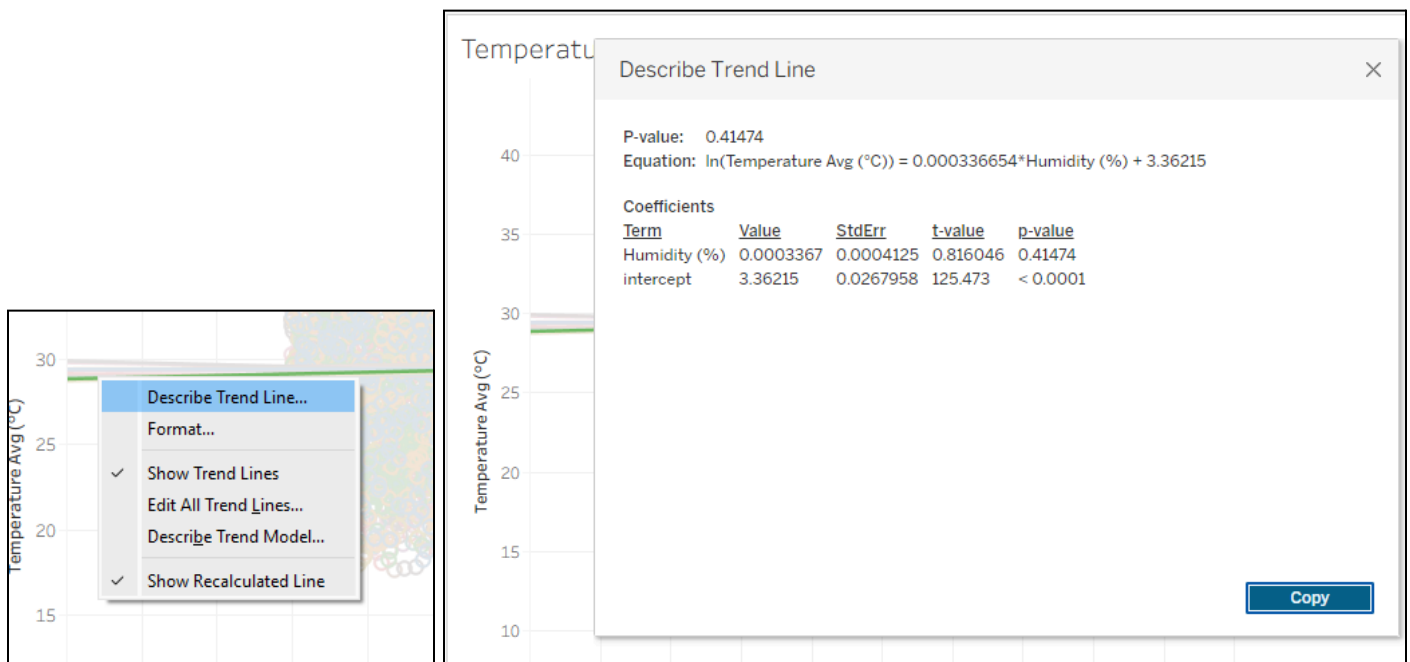


2. Compare Different Model Types (Polynomial & Logarithmic)

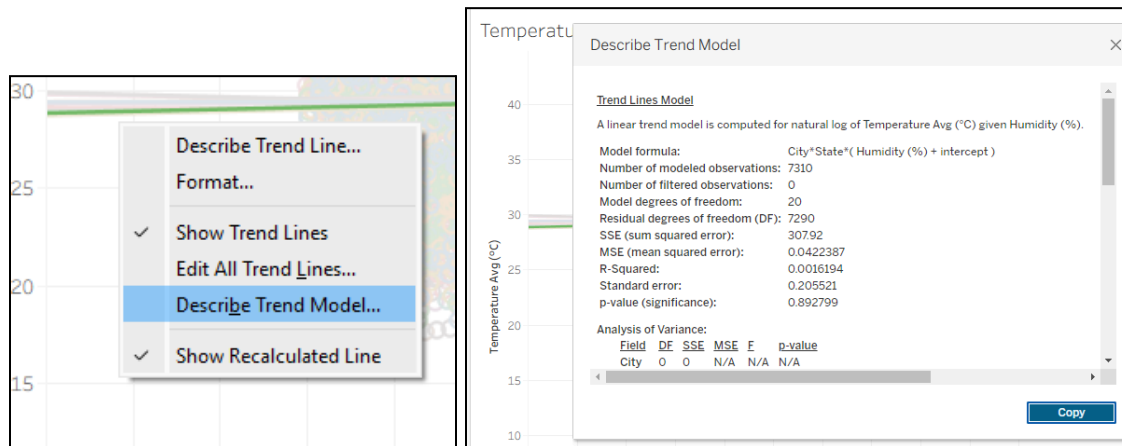


3. Access Full Statistical Summaries (Describe Model)

1. Right-click anywhere on the chart background or trend line and select Describe Trend Line...

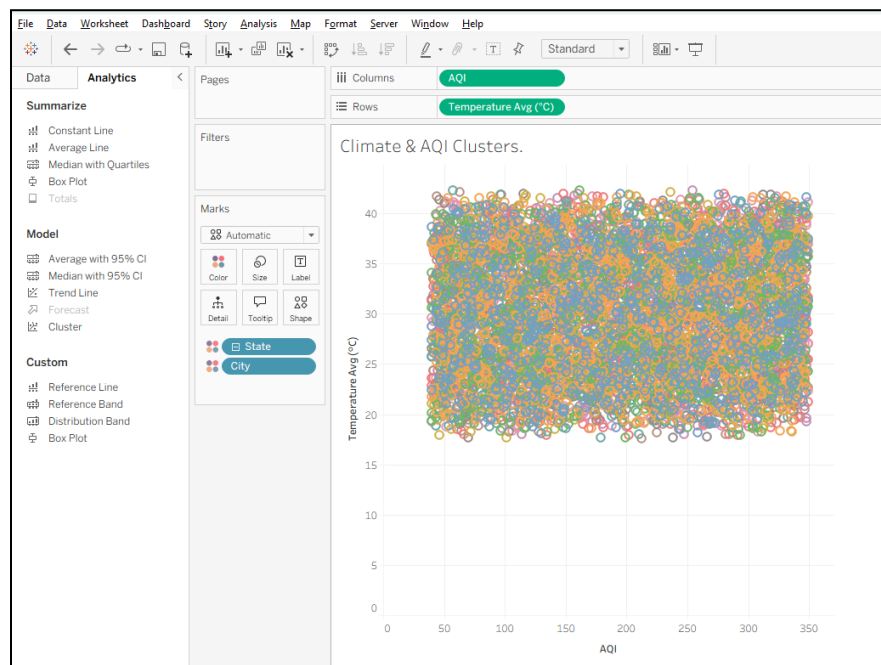


2. Next, right-click and select Describe Trend Model...

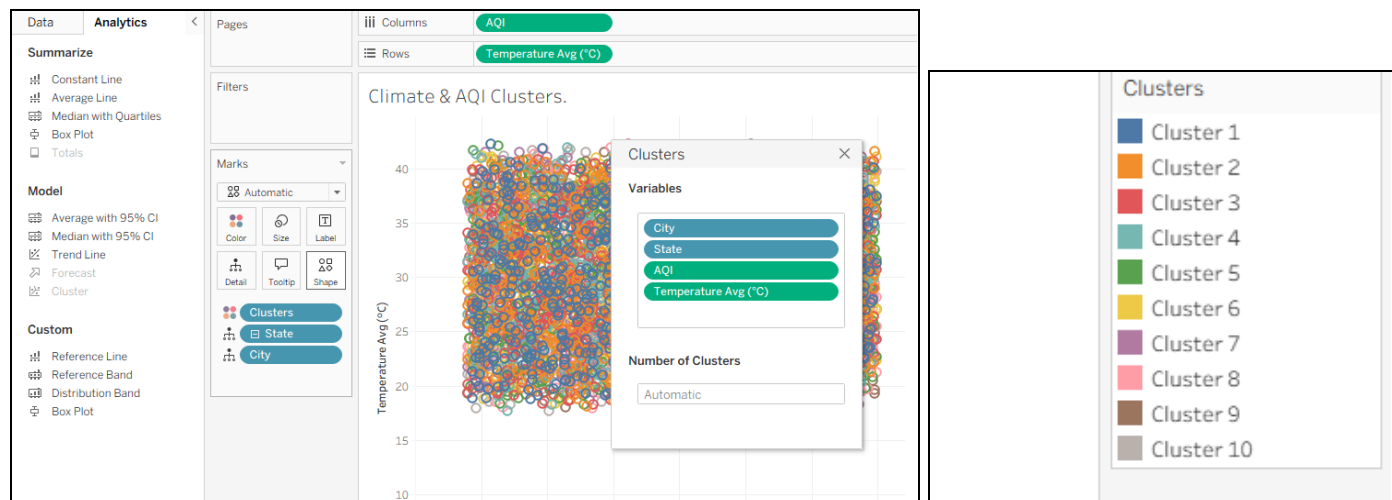


c) Perform clustering on datasets

1. Build a Multi-Metric Scatter Plot Matrix

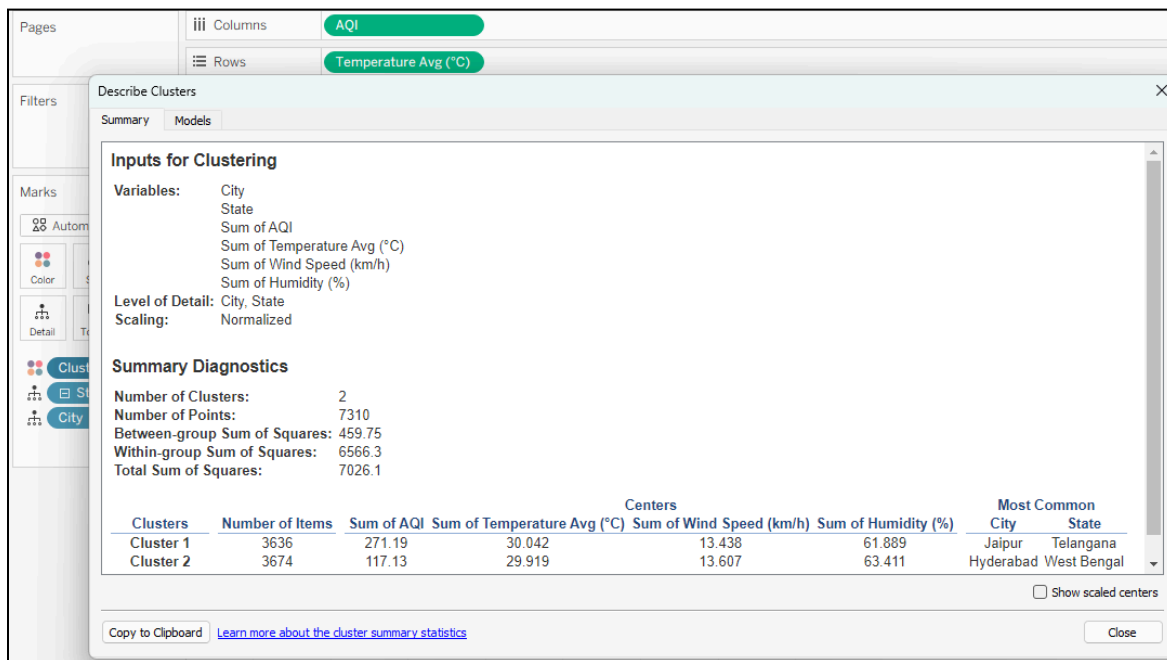


2. Apply Built-in Tableau Clustering



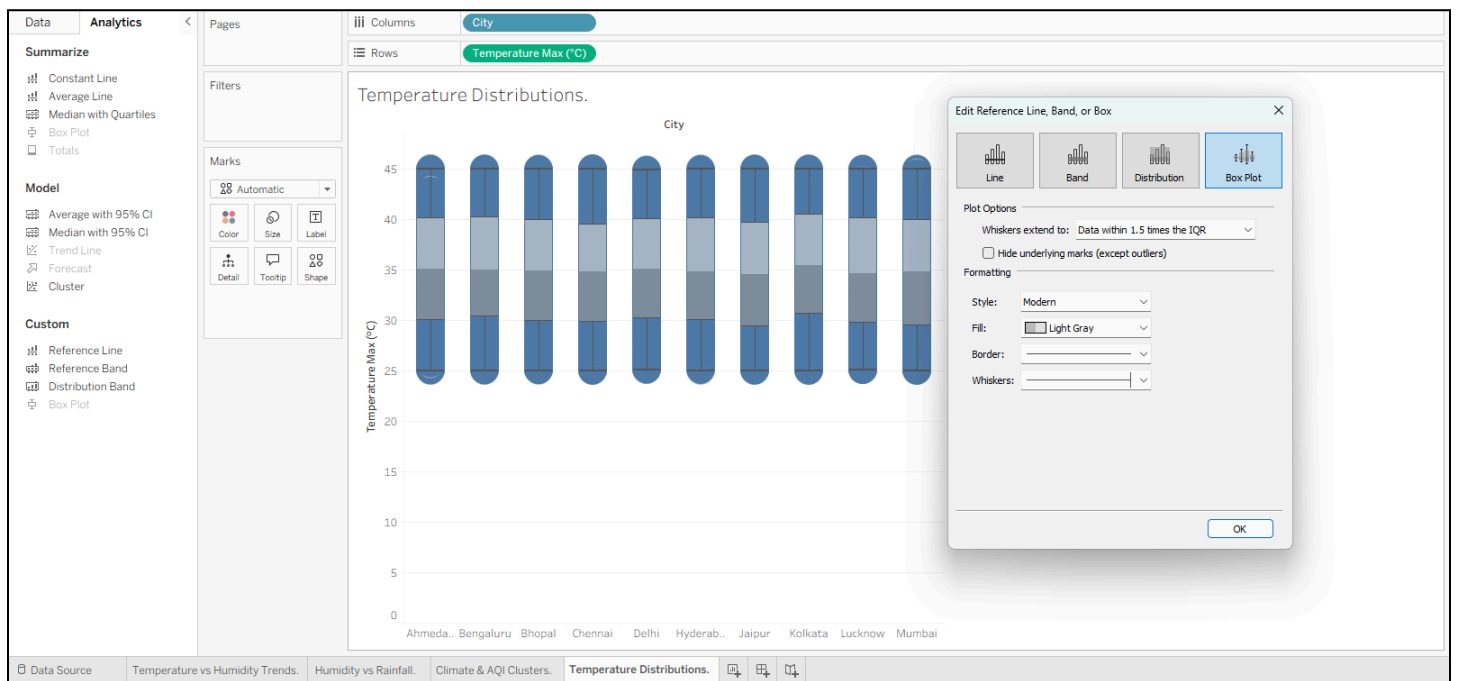
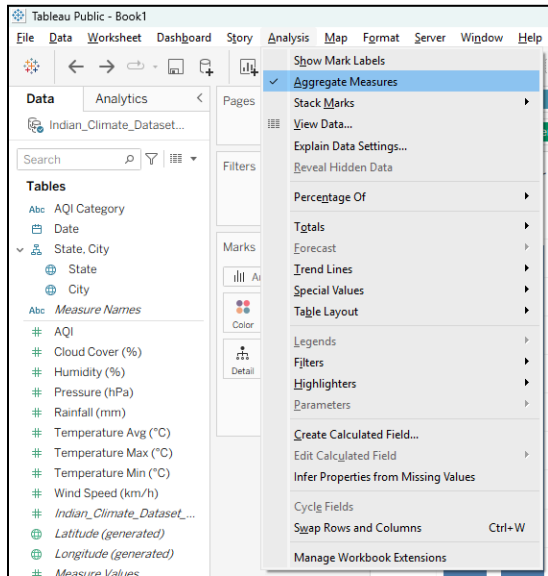


3. Analyze Cluster Profiles & Save as Dimension

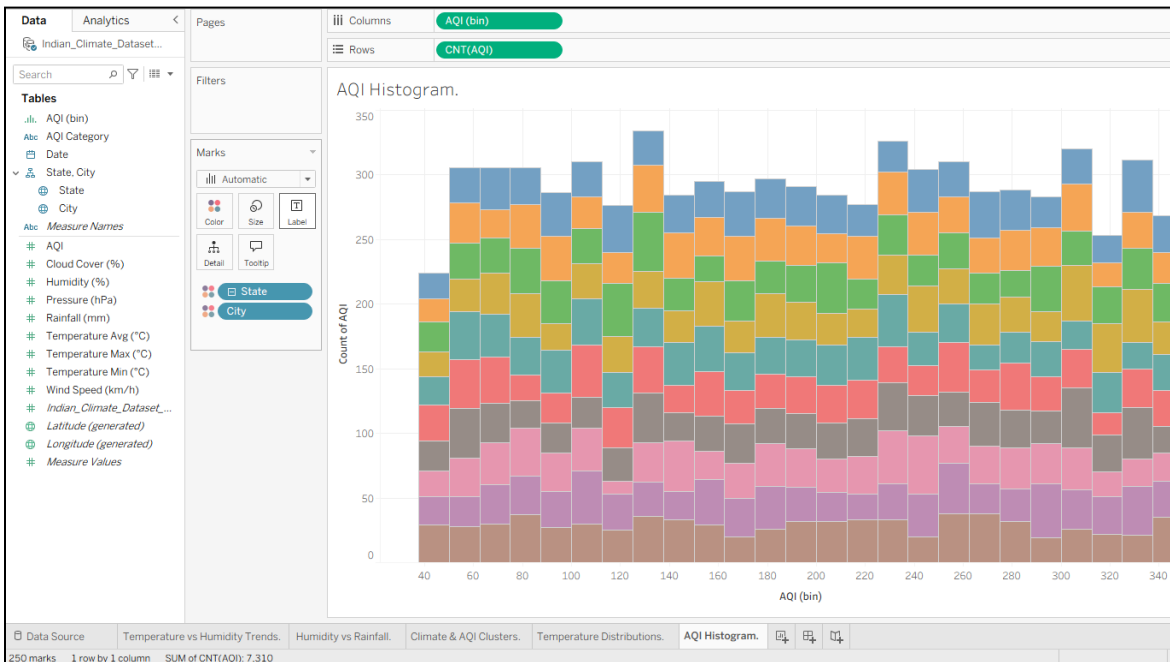
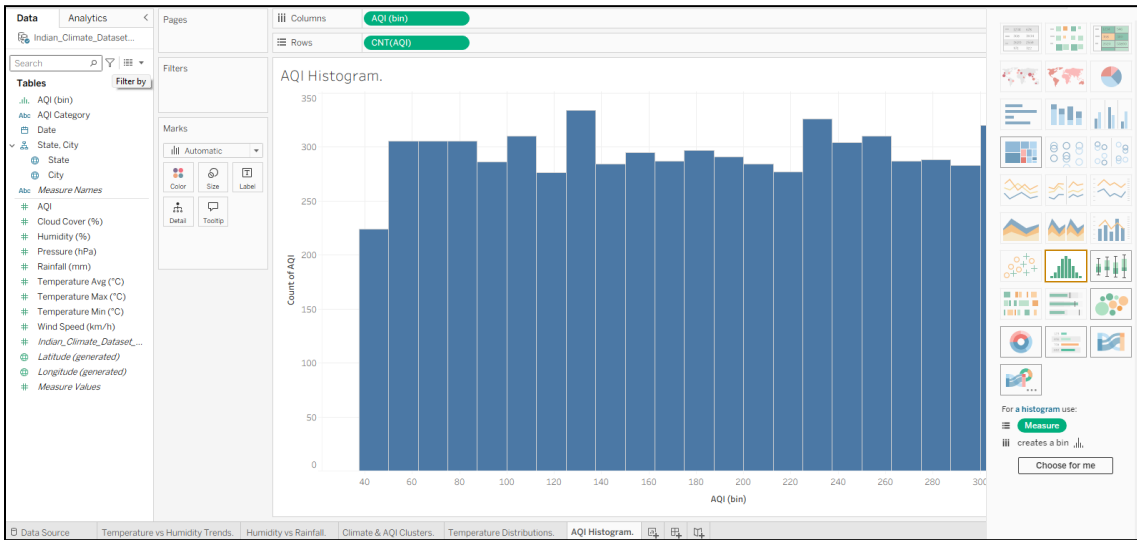


d) Analyze distributions using built-in tools

1. Create a Box Plot for Temperature Distributions by City

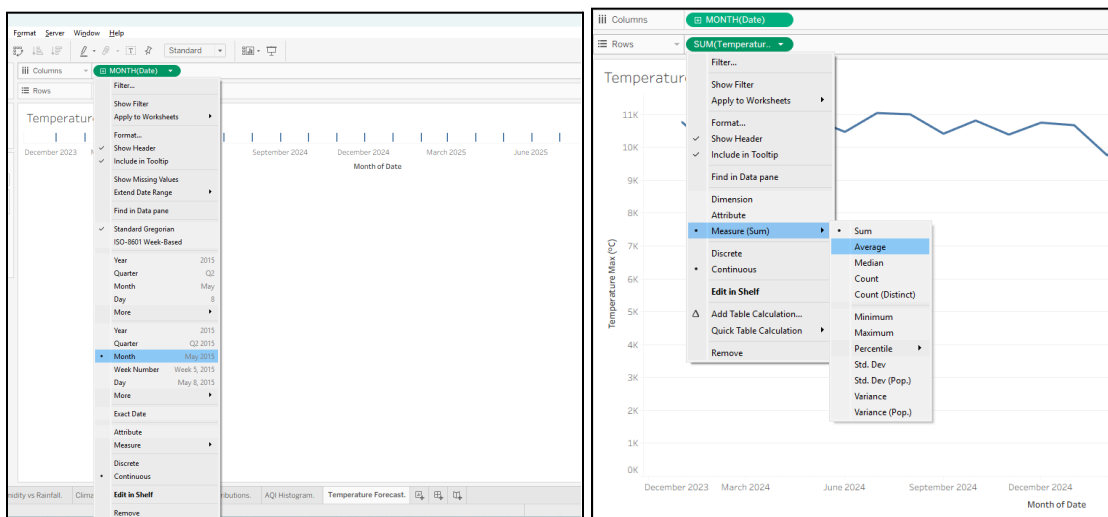


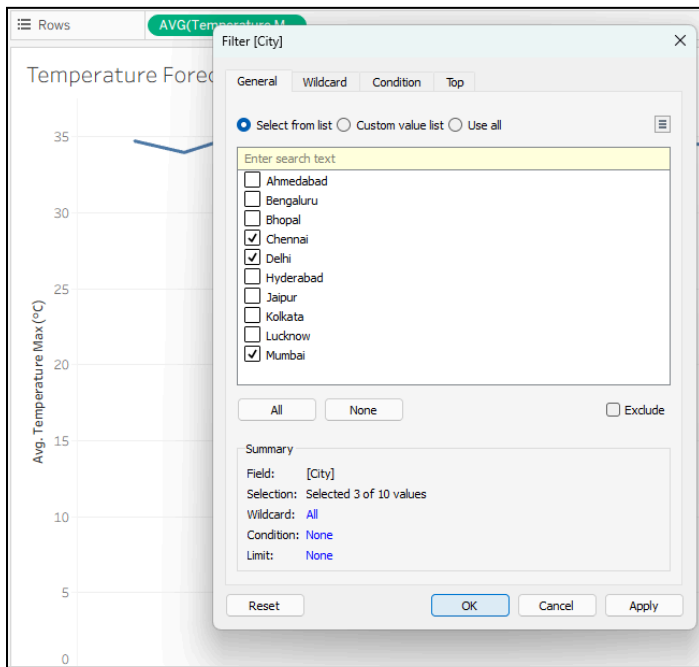
2. Build an AQI Histogram & Quantile/Percentile View



e) Apply forecasting techniques

1. Set Up a Time Series Line Chart

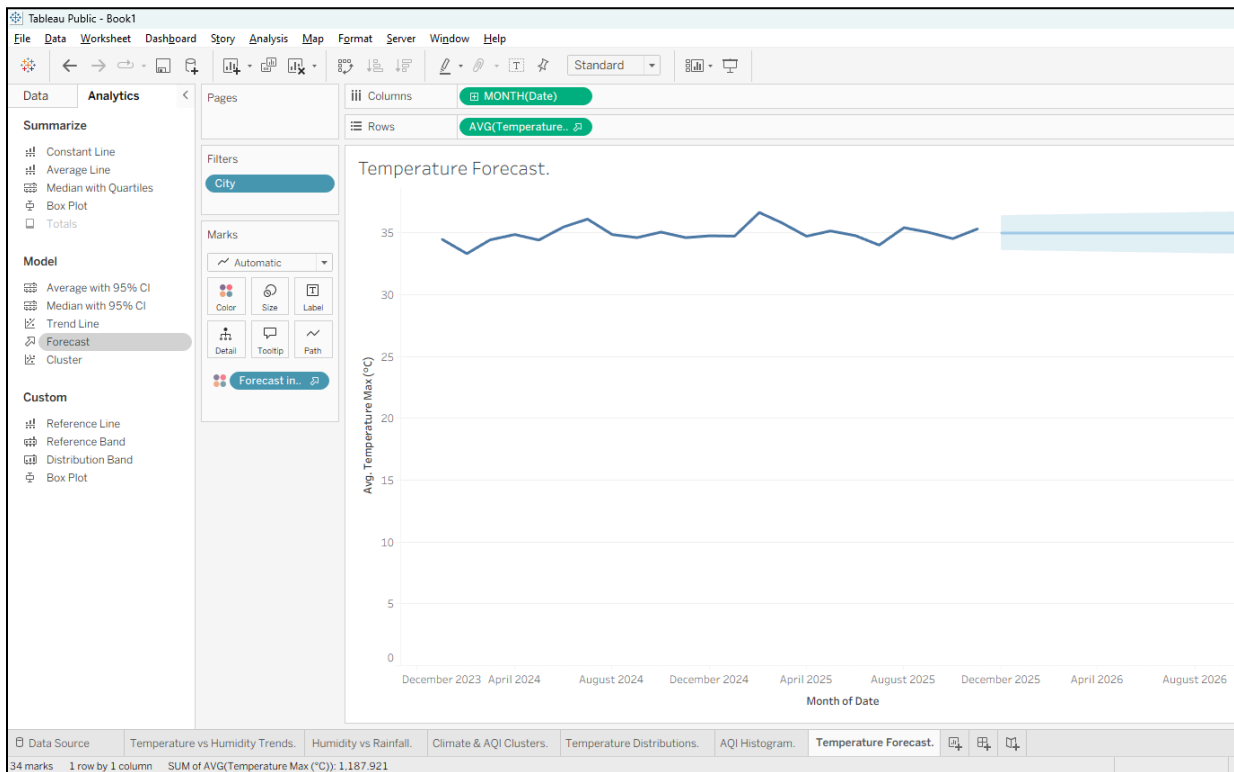




2. Enable Tableau Forecasting

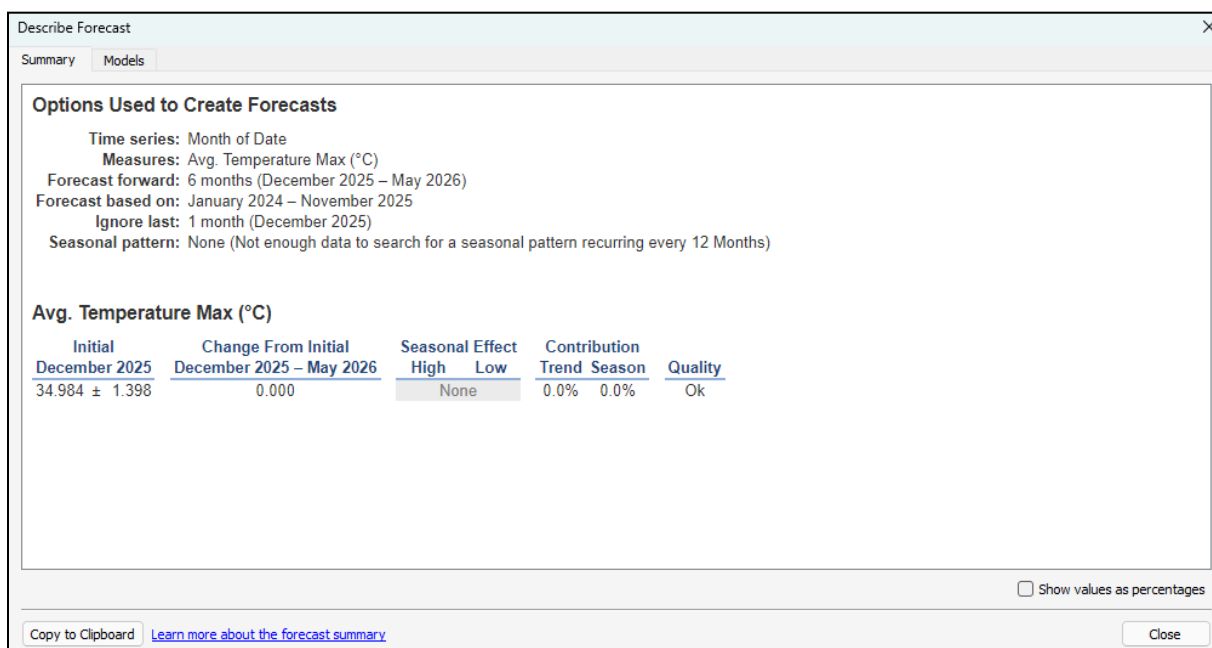
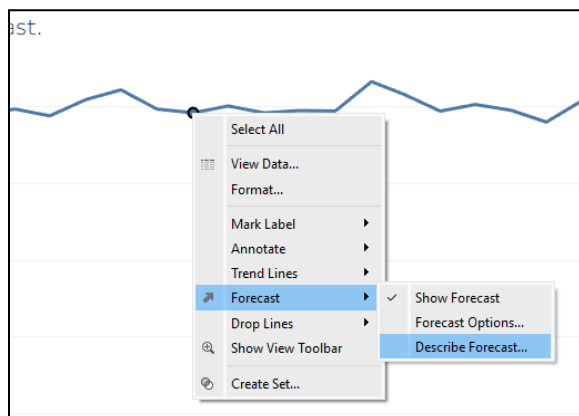
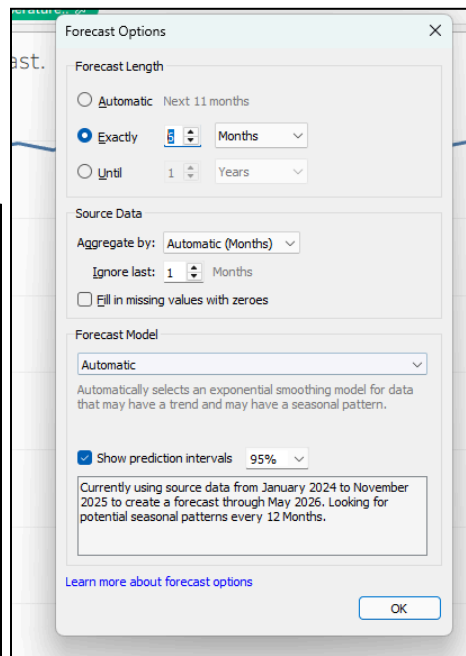
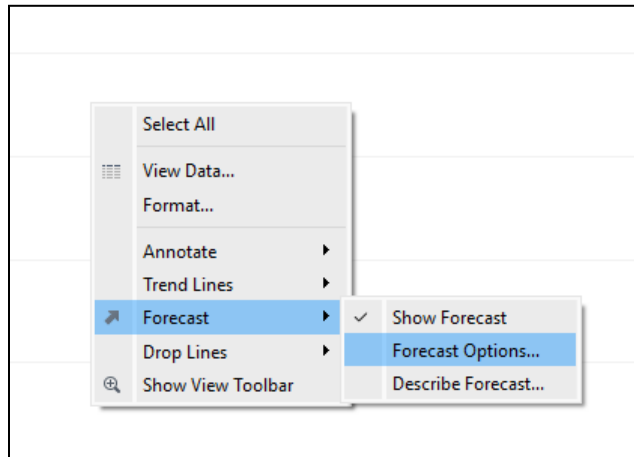
Switch to the Analytics Pane on the left sidebar.

Under Model, drag Forecast onto the canvas and drop it.



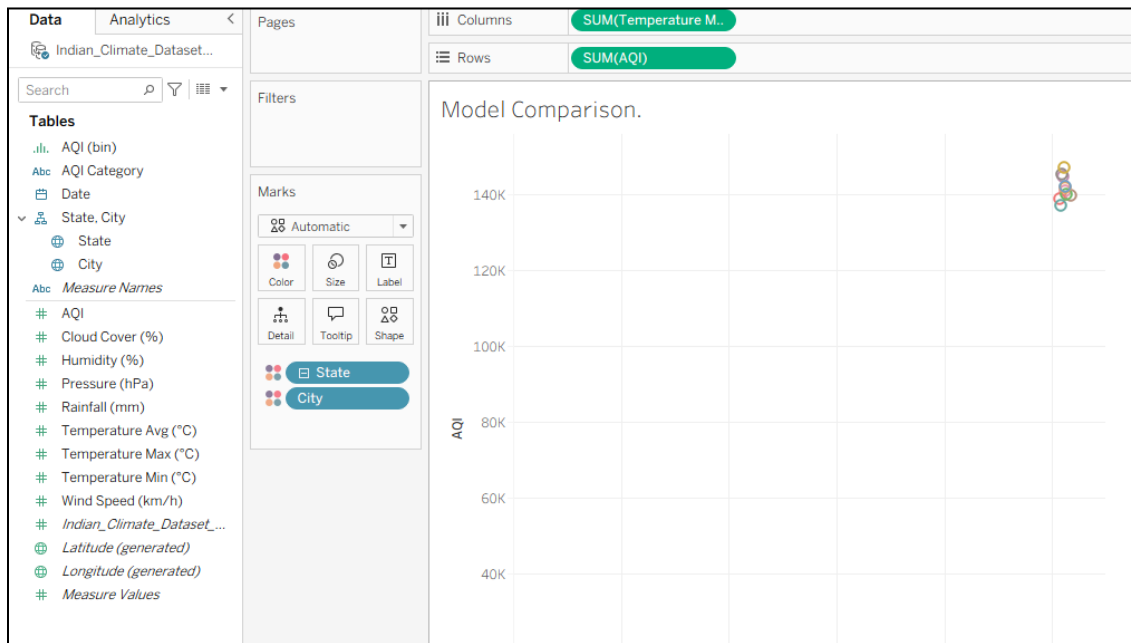
3. Customize & Analyze the Forecast

Right-click anywhere on the chart canvas select Forecast Forecast Options....

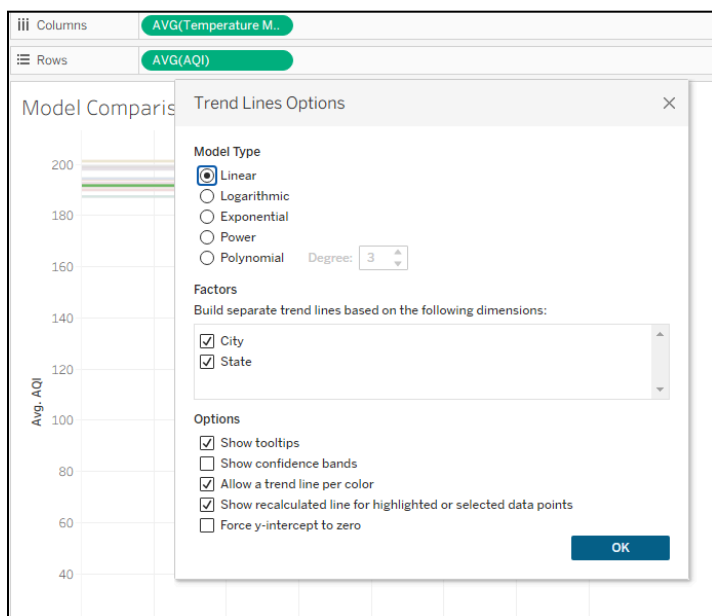
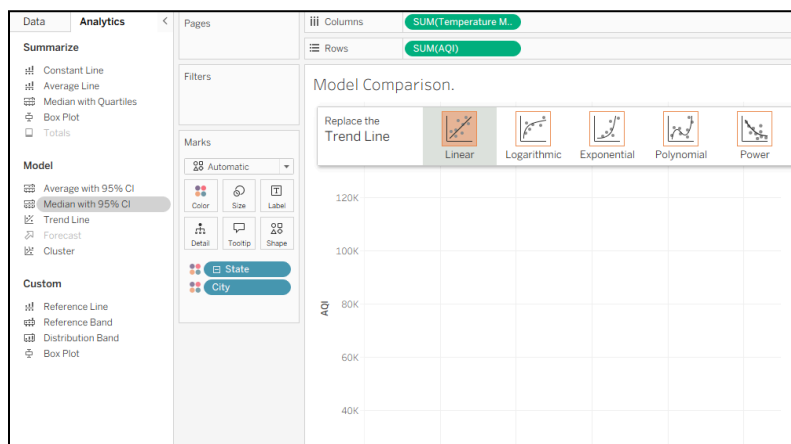


f) Compare different statistical models

1. Set Up the Comparison View

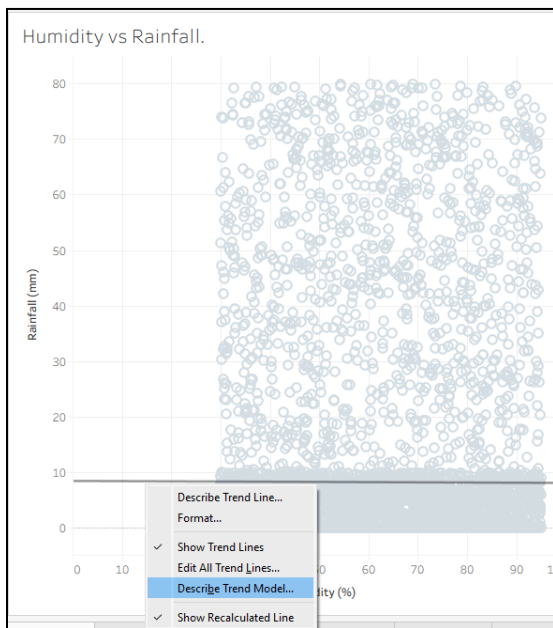


2. Add and Swap Statistical Models



g) Export and interpret statistical results

1. Export the Statistical Model Description



2. Interpret the Key Output Metrics

